LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 6 Test: Control and Coordination

Chapter / Topic: Control and Coordination

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SECTION A

Q1. Which of the following is a plant hormone? **[1]**

Answer: (b) Cytokinin

Q2. The gap between two neurons is called a: **[1]**

Answer: (c) synapse

Q3. The brain is responsible for: **[1]**

Answer: (d) all of the above

Q4. Which hormone regulates the blood sugar level? **[1]**

Answer: (a) Insulin (secreted by the pancreas)

Q5. The bending of a plant shoot towards light is called: **[1]**

Answer: (d) phototropism

Q6. **Assertion (A):** Iodised salt is advised in our diet. **Reason (R):** Iodine is needed by the thyroid gland to make thyroxin. **[1]**

Answer: (a). A deficiency of iodine can cause goitre.

SECTION B

Q7. What is a reflex action? Give two examples. **[2]**

Answer: A sudden, automatic response to a stimulus that happens without conscious thinking. Examples (any two): pulling the hand away from a hot object, blinking when a bright light falls on the eyes, sneezing, knee-jerk.

- Definition: 1
- Two examples: ½ + ½

OR, answer: The leaves fold up and droop when touched. Cells change shape by changing the amount of water in them, and there is no growth involved. The bending of a shoot towards light is a growth-related movement: cells on the shaded side grow longer because of auxin.

- Touch-me-not: water/cell shape change, no growth: 1
- Shoot: growth-related movement: 1

Q8. Why is adrenaline called the 'emergency hormone'? Give two of its effects on the body. **[2]**

Answer: It is secreted by the adrenal glands directly into the blood when we are scared or in danger, and it prepares the body to respond. Effects (any two): the heart beats faster, sending more oxygen to the muscles; blood supply to the digestive system and skin is reduced; the breathing rate increases.

- Secreted in emergency situations: 1
- Two effects: ½ + ½

SECTION C

Q9. Answer the following about a neuron: **[3]**

(i) At the dendritic tip (dendrites). [1]

(ii) From the dendrite to the cell body, and then along the axon to its end. [1]

(iii) At the end of the axon, the electrical impulse sets off the release of chemicals. These cross the gap (synapse) and start a similar electrical impulse in the dendrite of the next neuron. [1]

Q10. Name the plant hormone that: **[3]**

(i) Cytokinin [1]
(ii) Abscisic acid [1]
(iii) Auxin [1]

SECTION D

Q11. Answer the following: **[5]**

(i) (a) Pituitary gland: deficiency of growth hormone in childhood. (b) Pancreas: insulin not secreted in proper amounts. (c) Thyroid gland: thyroxin, due to iodine deficiency. [3]

- Each gland + hormone: 1

(ii) The timing and amount of hormone released must be precise, and feedback mechanisms regulate this. Example: when blood sugar rises, cells of the pancreas detect it and secrete more insulin. As the sugar level falls, insulin secretion is reduced. [2]

- Regulates timing/amount: 1
- Insulin example: 1

OR, answer: (a) Forebrain: the main thinking part, with areas for sensory information, voluntary actions, hunger and so on. Midbrain: controls many involuntary actions. Hindbrain: the cerebellum controls posture and balance, and the medulla controls blood pressure, salivation and vomiting. (b) The brain sits inside a bony box (the skull/cranium) and is surrounded by a fluid-filled balloon that acts as a shock absorber.

- (a) each region with function: 1
- (b) skull 1, fluid/shock absorption 1

SECTION E

Q12. A student placed a potted plant near a window for a week. The shoot bent towards the window, while the roots grew downwards. The student read that a… **[4]**

(i) Phototropism (positive). [1]
(ii) Auxin, made at the shoot tip. [1]
(iii) Auxin diffuses to the shaded side, where the cells grow longer than on the lit side, so the shoot bends towards light. Roots show positive geotropism (they grow downwards). [2]

- Shaded side cells elongate more: 1
- Positive geotropism: 1